

JUNAED HOSSAIN JIBON

Aspiring AI/ML Researcher — Deep Learning, Explainable AI, Computer Vision

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RESEARCH INTERESTS

Deep learning and explainable AI (Grad-CAM, LIME) for image-based classification; computer vision applications in agriculture and plant sciences; convolutional neural network architectures for real-world diagnostic and sensing problems.

EDUCATION

B.Sc. (Engineering) in Computer Science and Engineering Jul 2025

University of Asia Pacific, Dhaka, Bangladesh — CGPA: 3.55 / 4.00

Higher Secondary Certificate (Science) 2019

BAF Shaheen School and College, Paharkanchanpur — GPA: 4.08 / 5.00

Secondary School Certificate (Science) 2017

BAF Shaheen School and College, Paharkanchanpur — GPA: 4.86 / 5.00

RESEARCH EXPERIENCE

Undergraduate Thesis — Rice Leaf Disease Recognition Using Deep Learning and Explainable AI Jun 2024 – Jun 2025

Department of Computer Science and Engineering, University of Asia Pacific | Supervisor: Dr. Nasima Begum, Professor

- Designed and trained CNN-based image classification models (InceptionV3, VGG19, DenseNet201) to identify rice leaf diseases from field image data, applying transfer learning to improve performance on a limited dataset.
- Applied Grad-CAM to generate visual and local explanations of model predictions, improving interpretability for potential agronomic decision-making.
- Built the full pipeline in Python using TensorFlow/Keras — data preprocessing, augmentation, model training, evaluation, and explainability visualization.
- Work directly bridges computer vision, deep learning, and interpretable AI for agricultural applications.

SELECTED PROJECTS

Smart IoT-Based Dustbin — Arduino, ESP8266, Ultrasonic Sensor Jan 2024 – Jun 2024

- Led development of a touchless, IoT-enabled waste monitoring system with real-time level tracking, reducing manual inspection needs by ~30%.

Repair Anything — Full-Stack Repair Request Platform (Python, Django) Jul 2023 – Oct 2023

- Built a full-stack platform with authentication, booking, and admin dashboard; designed database models and CRUD operations for request management.

Computer Graphics Projects — C++ / OpenGL Jan 2025 – Jun 2025

- Solar system simulation and a real-time 2D game, applying texture mapping, transformation matrices, collision detection, and OOP-based architecture.

TECHNICAL SKILLS

- Machine Learning & Deep Learning: CNNs, transfer learning, Explainable AI (Grad-CAM), model evaluation, feature engineering, TensorFlow, Keras, Scikit-learn
- Programming: Python, C, C++, Java, SQL
- Tools: Jupyter Notebook, Git/GitHub, Docker (basic), PyCharm, Visual Studio
- Embedded Systems & IoT: Arduino, ESP8266, sensor integration

- Networking (professional experience): TCP/IP, VLANs, Routing & Switching, DNS/DHCP, ISP/POP operations

PROFESSIONAL EXPERIENCE

Assistant Network Engineer (POP In-Charge)

Feb 2026

Amber IT Limited, Bhairab, Kishoreganj

Embedded System Engineer (Internship)

Oct 2024

SoftRoot Bangladesh, Dhaka

HONORS & ACTIVITIES

- Dean's Honor List — Spring 2023, Spring 2021
- Participant, ICPC Asia Dhaka Regional Online Contest (2022, 2024)
- Volunteer, 45th ICPC World Finals Dhaka (2022) and Ada Lovelace National Girls Programming Contest (2022)

REFERENCES

Dr. Nasima Begum, Professor, Dept. of CSE, University of Asia Pacific — nasima.cse@uap-bd.edu | +880 1816 203075

Dr. Bilkis Jamal Ferdosi, Professor, Dept. of CSE, University of Asia Pacific — bjferdosi@uap-bd.edu | +880 1760 242388